

Referee Report on Round 13: Targets G2 and G3

Adversarial review of the D-discipline extraction and coverage walk

July 8, 2026

Overall verdict. Round 13 does not currently prove Theorem B(a) in D-form. The literal D2 propagation/verticalization clause is too strong for finite-width extraction: a fixed satisfiable concept forces infinitely many obligation-carrying vertical sources to be inserted into the same lower bag. The Shadow Amalgamation Lemma itself remains strongly supported under the horizontal-row discipline; the defect is on the completeness/extraction side that round 13 explicitly designated as G2. Status remains: strongly supported, not certified.

1 Executive verdict table

Target	Verdict	Finding
G2a	Fatal	Literal D2 fallback plus D4-style liveness is incompatible with any finite width bound. The witness concept C_{G2a} below is satisfiable, but for every prefix length m the D2 extraction of the standard vertical support chain forces the root edge bag to contain $m + 1$ distinct manifestations.
G2b	Fatal, derivative	The rows can be kept horizontal only by applying D2 fallback at every downward crossing of every remote source in the tower. Without fallback, D1 fails; with fallback, width fails.
G2c	Major	The “no third category” assertion is true only under the same globally omnidirectional D2 that causes G2a. Any width-preserving repair must replace Coverage by a relation-indexed lemma; the present proof cannot simply be reused.
G3a	Major	Under literal D2 the coverage walk is essentially trivial but unextractable. Under the needed weakening, the one-sentence dismissal of mixed walks is not a proof and lacks the invariant saying where propagation is allowed to stop and later restart.
G3b	Major, conditional	A closed three-point RCC5 network has $xPPz$, $zPPIy$, and $xPPy$. Thus a vertical endpoint relation can be supported by a mixed PP/PPI path, and D3 does not propagate an $\forall PP.D$ obligation to y . Literal D2 avoids this only by co-bagging x with y , which is exactly the source of G2a.

G3c	Sound under literal D2, unusable for extraction	I did not find an independent defect in Theorem 4.3’s row-realization argument under D1–D5. The problem is that the D2 hypothesis needed to exclude default pairs involving obligation sources is too expensive to extract at bounded width.
-----	--	--

2 G2a counterexample: verticalization versus width

The smallest literal-wording witness I found is

$$C_{\min} = \exists PP.\top \sqcap \forall PP.(\exists PP.\top \sqcap \forall PP.\exists PP.\top).$$

It already breaks round 13 if D2 is applied to every nonempty universal obligation set, including purely vertical universal obligations.

To remove the likely defense “vertical universals are meant to ride D3 and not cast shadows”, I used the following robust version in the machine check:

$$C_{G2a} = \exists PP.\top \sqcap \forall PP.(\exists PP.\top \sqcap (\forall PP.\exists PP.\top \sqcap \forall DR.A)).$$

The added $\forall DR.A$ makes every strict PP-successor of the root carry a genuinely horizontal universal obligation.

2.1 Semantic forcing

Let $a_0 \models C_{G2a}$. Choose a_1 with $a_0 PP a_1$. Since a_0 satisfies the outer $\forall PP$, the element a_1 satisfies $\exists PP.\top$, $\forall PP.\exists PP.\top$, and $\forall DR.A$. Choose a_2 with $a_1 PP a_2$. By $\text{Comp}(PP, PP) = \{PP\}$, $a_0 PP a_2$, so the outer universal at a_0 applies to a_2 as well. Inductively every finite prefix

$$a_0 PP a_1 PP a_2 PP \dots PP a_m$$

is forced, and every a_i with $i \geq 1$ has a nonempty horizontal obligation set because $\forall DR.A \in \text{tp}(a_i)$.

This is not an artefact of a chosen model. A finite cycle cannot close the chain: repeated PP composition would imply a nontrivial PP self-edge, contradicting strong equality/identity.

2.2 Why literal D2 forces unbounded bags

Use the standard edge-bag presentation of the selected PP support chain: B_i initially contains a_i and a_{i+1} , with adjacent bags sharing a_i or a_{i+1} as usual. Consider a remote source a_k with $k \geq 2$ and propagate its obligation shadow downward toward the root bag B_0 .

At the crossing from B_i to B_{i-1} , for $1 \leq i < k$, the continued separator occurrence is a_i . The true relation from source a_k to this separator is

$$a_k \text{ PPI } a_i,$$

because $a_i PP a_k$. A row entry $a_k \text{ PPI } a_i$ is illegal by D1 (horizontal rows). Therefore literal D2 says that the crossing is replaced by verticalization: a_k is inserted into the target bag as a live port. Repeating this over $i = k - 1, k - 2, \dots, 1$ inserts a_k into B_0 . Since this holds for every k , the root edge bag must contain

$$\{a_0, a_1, \dots, a_m\}$$

for the length- m prefix. The width requirement is therefore at least $m + 1$, for arbitrary m , for the fixed concept C_{G2a} .

The round-13 accounting by generation reasons (g1)–(g5), including $B_{r+1} \leq c(n)(1 + B_r)^2$, counts ports generated inside a bounded side context. It does not count the same lower bag receiving manifestations of unboundedly many independently generated remote obligation sources. These are manifestation copies, not new semantic elements, but width counts manifestations.

2.3 Machine check

I added `verification/wp18_g2a_verticalization_width_counter.py`. It checks three finite claims:

1. $C_{\min}$ and C_{G2a} are satisfiable according to the project oracle `cover_tree_tableau.check_sat`;
2. the complete PP-chain prefix network on $a_0, \dots, a_m$ is closed under the RCC5 composition table;
3. literal D2 fallback inserts all sources $a_1, \dots, a_m$ into the root edge bag.

The relevant output was:

```
C_min_literal = (ExPP.Top n AllPP.(ExPP.Top n AllPP.ExPP.Top))
oracle_sat_min = True closure_size = 12
C_G2a_robust = (ExPP.Top n AllPP.(ExPP.Top n (AllPP.ExPP.Top n AllDR.A)))
oracle_sat_robust = True closure_size = 18
```

finite prefix check under literal D2:

m	chain_closed	root_bag_size	expected	fallbacks
1	True	2	2	0
2	True	3	3	1
3	True	4	4	3
4	True	5	5	6
5	True	6	6	10
6	True	7	7	15
7	True	8	8	21
8	True	9	9	28
9	True	10	10	36
10	True	11	11	45

conclusion = PASS: root width grows unboundedly with the prefix

3 G2b: extracted horizontal rows

The tower above also diagnoses G2b. At each downward boundary, the extracted true row entry from source a_k to the continued separator a_i is PPI. Thus there are only two options:

1. keep the true row entry, violating D1; or
2. apply D2 fallback, inserting a_k as a live port in the target bag.

Round 13 chooses the second option, but G2a shows that the second option has unbounded simultaneous cost. Therefore the manuscript has not shown that “after saturation and threshold verticalization every shadowed pair has a horizontal true relation” within the claimed width bound.

This is stronger than the old overlap-amalgam stress case: no exotic side amalgam is needed. A single forced vertical tower with a harmless horizontal universal at each level is enough.

4 G2c: no third category

The current Coverage Lemma is stated for every occurrence y whenever $\Omega_x \neq \emptyset$. Under literal D2 this is almost tautological: x is live or shadowed in every bag reachable from its starting point, so a manifestation of y is either co-bagged with x or row-covered by x . However, that literal reading is precisely what fails G2a.

A width-preserving repair must allow propagation to stop at some vertical barriers. Once this is done, the trichotomy cannot remain occurrence-wide. It must become relation-indexed: for a fixed formula $\forall R.D \in \text{tp}(x)$ and a target y with $\rho_T(x, y) = R$, either the pair is co-bagged, or it is covered by a horizontal row if $R \in \{\text{DR}, \text{PO}\}$, or it is covered by a uniformly oriented vertical support chain if $R \in \{\text{PP}, \text{PPI}\}$. Pairs in a vertical barrier that are not in relation R should be allowed to remain undeclared for that particular obligation. The round-13 text does not state this weaker, relation-indexed lemma, and its existing proof does not establish it.

5 G3: the coverage walk

5.1 G3a: mixed walks

The proof of Lemma 6.1 says that mixed walks “end in cases (a)/(b) at the last horizontal manifestation’s bag family.” I could not make this sentence into a valid induction invariant. A mixed walk can switch from horizontal movement to vertical fallback, and after the fallback the source is live again; literal D2 then sends it in all directions, causing the G2a blow-up. If this propagation is weakened to preserve width, the proof needs an explicit rule saying which components are still covered and which are certified irrelevant for the current relation R .

Thus G3a is not an independent counterexample under literal D2, but it is not a repairable proof as written once G2a is fixed.

5.2 G3b: orientation bookkeeping

I added `verification/wp19_g3_mixed_orientation_probe.py`. It checks the closed three-point network

$$x\text{PP}z, \quad z\text{PPI}y, \quad x\text{PP}y.$$

The network is composition-closed because $\text{PP} \in \text{Comp}(\text{PP}, \text{PPI})$. For a source type containing $\forall \text{PP}.D$, D3 propagates D and $\forall \text{PP}.D$ from x to z along $x\text{PP}z$, but it does not propagate from z to y along $z\text{PPI}y$. The output was:

```
B0={x,z}, B1={z,y}, overlap={z}
relations: x PP z, z PPI y, completion x PP y
composition_closed = True
D3_from_AllPP_D_at_x gives D flags: {'x': False, 'z': True, 'y': False}
D3_from_AllPP_D_at_x gives AllPP.D flags: {'x': True, 'z': True, 'y': False}
endpoint_relation_x_y = PP
conclusion = PASS: mixed PP/PPI path can have vertical endpoint; uniformity needs an extra inv
```

This is not a valid D-abstraction under literal D2, because D2 would verticalize x across the first boundary and make x, y co-bagged in the second bag. But that observation is exactly the problem: the orientation claim in case (c) is safe only because D2 is doing expensive global co-bagging. It is not a consequence of the composition table or patchwork alone.

5.3 G3c: interaction with Theorem 4.3

I found no direct defect in the row-realization part of Theorem 4.3 under the literal D-discipline. D1, D5, and the extended-family patchwork proof do make declared horizontal row values literal values of ρ_T . Also, if D2 is read literally, default DR pairs cannot involve obligation-carrying sources, because those sources have been propagated everywhere.

The caveat is again extraction: the literal D2 hypothesis needed for this argument is not finite-width extractable. After a width-preserving repair, the “default pairs involve only obligation-free sources” statement will need to be replaced by a relation-indexed irrelevance statement.

6 Steps checked and found sound

The following pieces survived this targeted review.

1. I reran WP17. The four finite RCC5 facts used by the Shadow Amalgamation Lemma passed; 400/400 random D-discipline tree configurations were extended-bag closed and globally realizable; the negative control still triggered.
2. I reran WP16 in repaired mode. The listed referee diagnostics and the random sweep agreed with the cover-tree tableau oracle. This supports the acceptance-level intuition but does not test the finite-width extraction failure above.
3. I reran WP14. The two-bag patchwork checks, finite tree-patching stress test, and strong-EQ non-forcing check passed.
4. I did not find a new defect in the horizontal Shadow Amalgamation Lemma itself. The D1 horizontal-row restriction plus D5 pair coherence are exactly what make the extended-bag closure proof work.
5. The D3 propagation rule is sound for uniformly oriented vertical chains: $\text{Comp}(\text{PP}, \text{PP}) = \{\text{PP}\}$ and $\text{Comp}(\text{PPI}, \text{PPI}) = \{\text{PPI}\}$.

7 Recommended round-14 repair direction

Round 14 should not try to save literal D2. The repair should split obligation coverage by relation and replace omni-directional port verticalization with a bounded horizontal-frontier mechanism.

Concretely, define horizontal obligations

$$\Omega_x^H = \{(R, D) \in \Omega_x : R \in \{\text{DR}, \text{PO}\}\}$$

and vertical obligations Ω_x^V analogously. Vertical obligations should be handled by D3 along certified uniformly oriented support chains, not by long-range shadows. Horizontal shadows should propagate only across open horizontal frontiers. When a crossing is known to enter a vertical cone of the source, the automaton should record a finite vertical-barrier certificate rather than inserting the source as a port. If a later side context exits the cone and creates a possible DR/PO target for the source, the certificate must reopen a horizontal row there.

The corresponding Coverage Lemma should be restated as:

For every $\forall R.D \in \text{tp}(x)$ and every y with $\rho_T(x, y) = R$, the pair (x, y) is either co-bagged, row-covered by a horizontal row when $R \in \{\text{DR}, \text{PO}\}$, or connected by a co-bagged uniformly oriented vertical chain when $R \in \{\text{PP}, \text{PPI}\}$.

This is a proposed repair path, not a completed proof. The new proof obligation is to show that vertical-barrier certificates are locally checkable, width-bounded, and sufficient to reopen horizontal rows exactly when a horizontal target can occur.

8 Final recommendation

Reject round 13 as a complete repair of Theorem B. Keep the Shadow Amalgamation Lemma as a strongly supported component, but withdraw the claimed D-form extraction theorem and Coverage Lemma until D2 is replaced and the relation-indexed coverage proof is supplied.